

Triple-Band Terahertz Metamaterial Ultra-Sensitive Absorber for Multi-Class Cancer Cell Detection

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Abstract— This study introduces a meticulously optimized, Tri-spectral THz meta-absorber designed for high-resolution spectroscopic interrogation of cancer cell detection. The device architecture employs copper-based square-ring resonators atop a cost-efficient FR-4 dielectric substrate, configured within a multilayer stack to support distinct resonance modes at 0.717, 1.368, and 3.535 THz. Geometric parameter tuning via a genetic algorithm facilitates enhanced field confinement and spectral selectivity, yielding high absorption magnitudes approaching 99.99%. The absorber exhibits pronounced sensitivity to minute dielectric perturbations, achieving sensitivity to the refractive index peaking at 7357.14 GHz/RIU, alongside elevated quality factors and narrow linewidths. Electromagnetic simulations validate its robustness against angular incidence and mechanical deformation, underscoring its applicability in dynamic or flexible sensing environments. The proposed platform demonstrates strong potential for label-free, non-invasive differentiation of malignant cellular structures, thereby offering a promising modality for next-generation terahertz biosensing in oncological diagnostics.

Keywords—Metamaterial absorber; multiband THz, perfect absorption; cancer cell detection; photonic biosensor; THz sensing.

I. INTRODUCTION

The development of sensitive terahertz devices for detecting biomolecules has become a major focus of research, given their importance in biomedical applications. Metamaterial-based perfect absorbers (MPAs), which are periodic engineered structures specifically designed to exhibit distinctive electromagnetic characteristics, enable precise manipulation of wave absorption. In the last decade, absorbers based on metasurfaces have demonstrated remarkable potential by enhancing sensing, imaging, and diagnostic technologies. These advancements have, in turn, propelled innovation across numerous scientific and technological domains [1]. Recent research has explored terahertz (THz) metamaterial biosensors for cancer detection. These sensors take advantage of THz waves sensitivity to polar elements and the changes in dielectric constants in cancerous tissues [2]. In contrast, several studies utilize resonance frequency shifts induced by changes in the refractive index of biological samples for the detection of cancerous cells [3, 5, 7, 8]. The proposed designs achieve high absorption rates, ranging from 97% to 99.43%, at terahertz frequencies [3, 5, 7]. They also

demonstrate high sensitivity, with values reported between 360 GHz/RIU and 1447 GHz/RIU [3, 5]. Another dual-band sensor by M. N. Hamza et al., operate at 0.78 and 0.904 THz, demonstrated over 99.5% absorption for non-melanoma skin cancer diagnostics [4]. A. R. Elhelw et al. proposed a triple-band sensor for skin, blood, and breast cancer detection achieved high sensitivity of 2050 GHz/RIU and a quality factor of 55.34 in the 4.25–4.75 THz range [6]. These sensors have been applied to various cancer types, including brain cancer [7] and multiple cell types such as basal, breast, Jurkat, and cervical cells [8]. Additionally, some designs exhibit polarization insensitivity, enhancing their versatility [5, 7]. These advancements show promise for early and efficient cancer detection in biomedical applications. Polarization angle, field distributions, and absorption peaks demonstrate the potentiality of sensors for early cancer detection and tissue distinction.

This work highlights a compact Tri-resonant THz meta-absorbers operating in the THz range, specifically designed and simulated for the detection of different kinds of anomalies cell, achieving near-unity absorption at 0.717, 1.368, and 3.535 THz. Geometric optimization enhances field localization, and the symmetric design ensures polarization and angle insensitivity. With peak absorptions of 99.99%, 99.95%, and 99.99%, the design is validated through simulations. Its novelty lies in combining a low-cost FR-4 substrate with high sensitivity and multi-band absorption for precise, label-free cancer detection.

II. DESIGN METHODOLOGY

The functionality of terahertz Tri-band metamaterial absorber for identification of malignant cells illustrated in Fig. 1. The system comprises three key stages comprising sample collection and preparation of biological specimens; interaction of incident THz radiation with the metasurface structure incorporating a multilayer unit cell and spectral analysis of the reflected signal, enabling detection of resonance shifts corresponding to distinct cancer cell phenotypes.

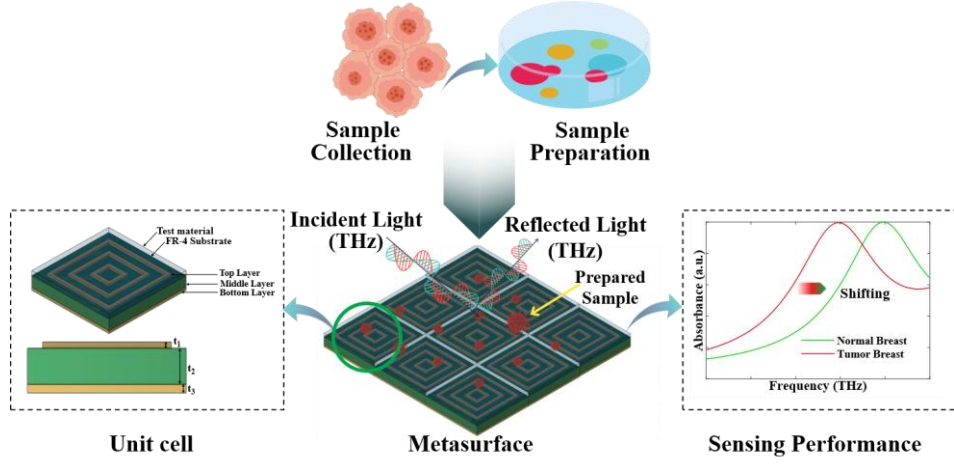


Fig. 1: Schematic representation of the proposed terahertz metamaterial absorber based on biosensing framework.

III. DESIGN ANALYSIS

A. Unit Cell Design

The suggested design went through several rounds of evaluation, during which several substrates, including Quartz, FR-4, and MgF₂ were considered. The refined design, employing a FR-4 substrate, ended up with dimensions of $53 \times 55 \times 7.8 \mu\text{m}^3$. The structure exhibits copper coverage on both the upper and lower layers, with the upper layer distinctly featuring a square ring-shaped copper arrangement. The refined outcomes, depicted in Fig. 3 and Fig. 4, are examined comprehensively in the following sections.

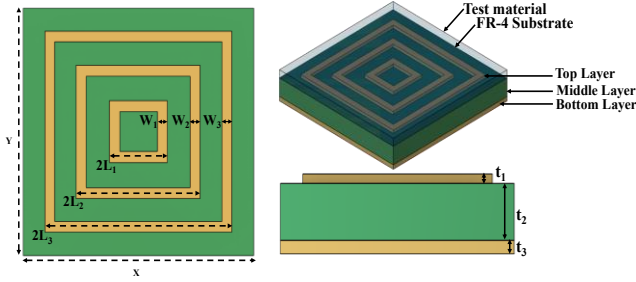


Fig. 2: 3D view of the unit cell with precise geometric measurements for the proposed terahertz metamaterial absorber

B. Parameter Optimization

The metamaterial (MTM) absorber design features a periodic unit cell at its core, as shown in Fig. 2. To optimize absorption, the genetic algorithm (GA) in CST Microwave Studio was used, with Fig. 4(f) illustrating the process. The algorithm systematically modifies structural parameters to improve efficiency within the designated frequency bands, executing 25 iterations with a mutation rate of 65% and an anchorage length of 10%. The spectral response was assessed using typical receiving electromagnetic waves. The optical properties, defined by transmittance (T), absorption (A), and reflectance (R) were calculated from scattering parameters S_{11} and S_{21} [2, 9]. Absorption depends on energy conservation and is determined by $R(\omega) + T(\omega) + A(\omega) = 1$, maintaining metamaterial energy balance.

$$S_{11} = \frac{\sqrt{P_r}}{\sqrt{P_i}} \quad (1)$$

$$S_{21} = \frac{\sqrt{P_t}}{\sqrt{P_i}} \quad (2)$$

$$R(\omega) = \frac{S_{11}^2}{S_{11}^2} \quad (3)$$

$$T(\omega) = \frac{S_{21}^2}{S_{21}^2} \quad (4)$$

Where, P_i =Incident Power, P_r =Reflected Power, and P_t =Transmitted Power respectively.

To investigate the effects of material and dimensional variations, five distinct simulation sets were conducted, as shown in Fig. 4. These simulations provided valuable insights into the resonant behavior of the absorber. Materials such as FR-4, MgF₂, and Quartz was chosen to investigate the impact of dielectric properties on performance [10]. The material choice significantly impacted the absorption behavior. Fig. 3(a) indicates that both Quartz and MgF₂ demonstrated absorption values greater than 86% at varying frequencies: 0.755 and 0.844 THz for Quartz 0.647 and 1.22 1 THz for MgF₂. Conversely, FR-4 exhibited greater absorption, achieving 99.99% at frequencies of 0.717 and 3.535 THz. The second resonance for FR-4 exhibited an absorption rate of 99.95% at 1.368 THz, whereas MgF₂ and Quartz recorded absorption levels of around 70–74%. The variations in material properties, as detailed in Table I, had a significant impact on the resonant behaviour. Fig. 3(b) demonstrates the impact of varying the metal element (Gold, Copper, Aluminium) on performance, resulting in a triple-band response [10]. Copper demonstrated exceptional conductivity, resulting in the most favourable spectral response, with an absorption rate of 99.99%. Furthermore, the integration of a FR-4 substrate with a copper plate produced three unique absorption peaks at frequencies of 0.717, 1.368, and 3.535 THz, corresponding to a thickness of $t_1 = 0.24 \mu\text{m}$. The results underscore the importance of material selection in enhancing performance across various frequency bands. Fig. 4 (a-c) highlights the significance of resonator lengths (L_1 , t_1 , W_1) in the tuning of absorption peaks. The first resonator length, L_1 , which varied from $5 \mu\text{m}$ to $9 \mu\text{m}$, had a notable impact on the third resonance frequency. Adjustments to the second resonator length, L_2 (16 – $20 \mu\text{m}$), resulted in significant shifts in the second resonance frequency, highlighting the system's sensitivity to changes in dimensions. Changes in the outer resonator length L_3 influenced the first absorption frequency, resulting in frequency shifts within the 0.6 – 1 THz range. The optimal absorption of 99.99% was recorded at $L_3 = 25.5 \mu\text{m}$, with L_1 set at $7 \mu\text{m}$ and L_2 at $18 \mu\text{m}$. Absorption values of 99.73% and 99.99% were observed at resonator lengths of $L_1 = 7 \mu\text{m}$,

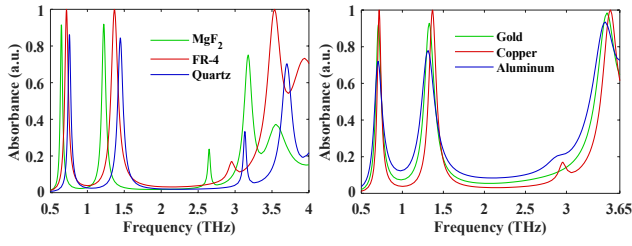


Fig. 3: Optimization of (a) substrate material and (b) radiating element geometry for absorption and sensing.

TABLE I: MATERIAL PROPERTIES

Metals	Plasma frequency ($\omega_p \times 10^{15} \text{ S}^{-1}$)	Collision frequency ($\nu_c \times 10^{15} \text{ S}^{-1}$)
Gold	13.8	0.11
Aluminium	22.9	0.92
Copper	13.4	0.14

as illustrated in Fig. 4 (a), with L_2 set at $18 \mu\text{m}$ and L_3 at $25.5 \mu\text{m}$. These findings demonstrate how resonator dimensions affect frequency-dependent absorption performance. Increased thickness in slot, substrate, and ground plane in 3D material layers increases antenna reflection co-efficient (t_1 (0.16 - $0.28 \mu\text{m}$) in Fig. 4 (b), t_2 (7.2 - $8.4 \mu\text{m}$), and t_3 (0.33 - $0.41 \mu\text{m}$), but does not affect resonance frequency. This highlights the precision needed in slot and resonator designs. Optimal characteristics were achieved with FR-4 thickness and width set to $t_2=7.8 \mu\text{m}$ and $y=55 \mu\text{m}$. The optimization of the square ring widths (W_1, W_2, W_3) within the THz meta-absorber structure was found to have a significant impact on the reflection coefficient at the targeted resonance frequencies. Systematic variation of these widths revealed that the optimal performance was achieved with $W_1 = 2.05 \mu\text{m}$, $W_2 = 2.5 \mu\text{m}$, and $W_3 = 1.65 \mu\text{m}$. Specifically, these dimensions resulted in reflection coefficients of -42.78 , -34.97 and -42.59 dB at 0.717 , 1.368 and 3.535 THz respectively for W_1 . For W_2 and

W_3 it creates distinct three peaks values of -42.47 , -54.59 , -39.53 dB and -42.47 dB, -54.51 dB, and -39.20 dB, respectively at respective resonance peaks. These findings underscore the critical role of geometric parameter tuning in optimizing absorption characteristics, with the W_2 configuration exhibiting the most pronounced reflection suppression at 1.368 THz. Furthermore, the consistency of the resonance frequencies across different width configurations indicates robust structural stability, which is essential for practical sensing applications in the terahertz regime. Fig. 4 (c) presents the corresponding S-parameter plot. The findings highlight the complex interplay among structural dimensions, material selection, and electromagnetic responses in metamaterial absorbers. Fig. 4 (d) illustrates that increasing the bending radius from $0 \mu\text{m}$ to $70 \mu\text{m}$ and $100 \mu\text{m}$ significantly enhances absorbance at the three primary resonant frequencies: 0.717 , 1.368 , and 3.535 THz, with the $100 \mu\text{m}$ radius exhibiting the highest and most pronounced peaks. This improvement is due to bending-induced changes in the optical path and effective refractive index, which enhance field confinement and mode coupling. Such deformation-driven tunability highlights the absorber's potential for high-performance, flexible terahertz sensors, particularly in sensitive biomedical applications like cancer cell detection. Fig. 4 (e) illustrates the interplay between incidence angle (0° - 90°) and frequency (0.5 - 3.5 THz) in a THz spectroscopy system, with color intensity (1.0 - 0.0) likely representing signal efficiency. At lower frequencies (0.5 - 2 THz), signal efficiency remains stable across angles, suggesting optimal conditions for biosensing applications. However, higher frequencies (>2 THz) exhibit significant attenuation, particularly at steep angles ($>70^\circ$), likely due to increased scattering or reflection losses.

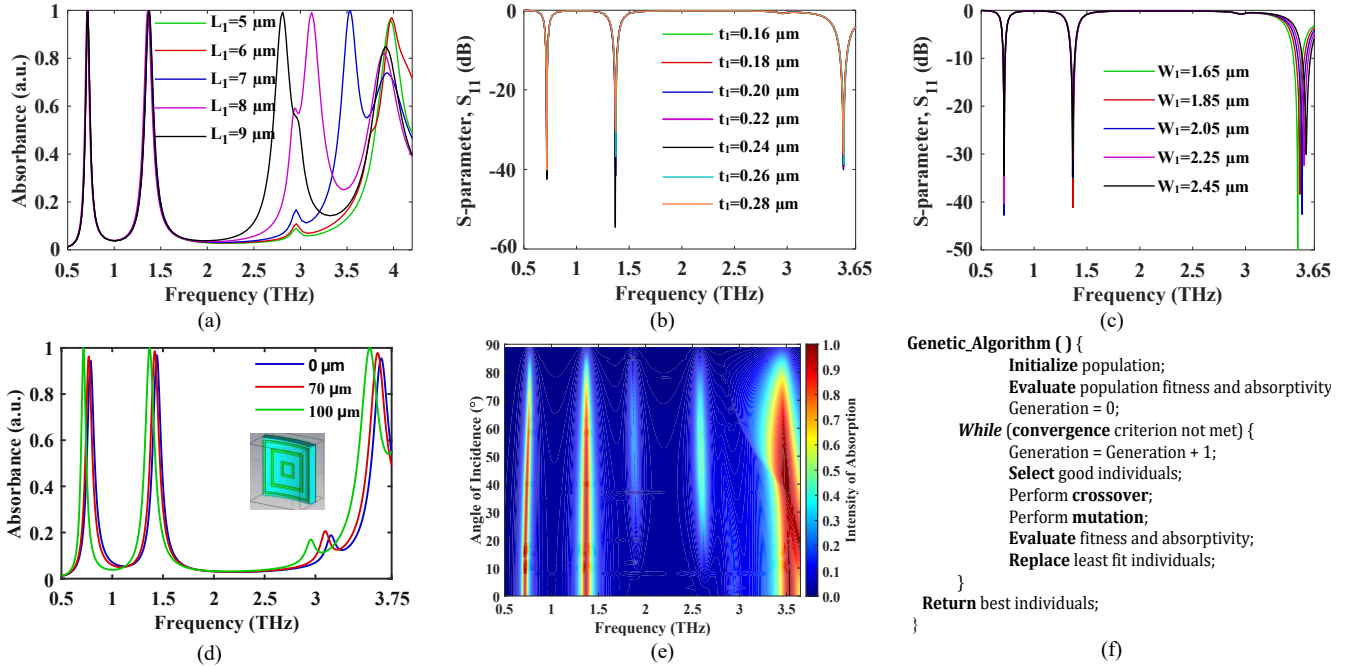


Fig. 4: Parameter optimization of the metamaterial absorber: (a) effect of length variation, (b) influence of substrate thickness, (c) impact of resonator width, (d) responsiveness to change in bending radius, (e) interaction with incidence angle on absorption performance and (f) pseudo code of design optimization.

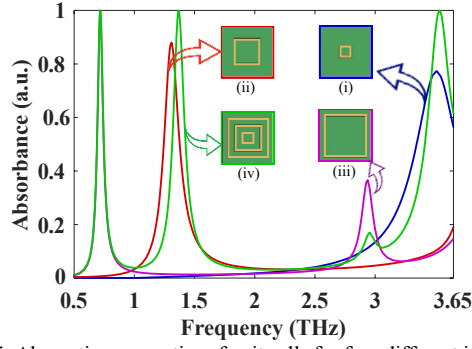


Fig. 5: Absorption properties of unit cells for four different internal structures of the resonator

This data underscores the importance of selecting angles (0° – 60°) and lower THz frequencies to maximize sensitivity for detecting refractive index variations between healthy and cancerous cells, a critical factor in the proposed biosensor's design for accurate cancer diagnostics. Four metamaterial absorber concepts with different square ring shapes demonstrate absorption capability in the 0.7–3.6 THz range in Fig. 5. The findings indicate that the dimensions and intricacy of the square rings substantially influence absorption. The basic design, featuring a small central square ring (i), exhibits a peak at 3.51 THz, though it demonstrates restricted tunability. The second design, featuring a larger middle square ring (ii), achieves a peak absorption of 87.97% at 1.31 THz. The third configuration, featuring the largest outer square ring (iii), attains near-perfect absorption ($\sim 99.99\%$) at 0.717 THz. The final design, featuring concentric square rings (iv), attains approximately 99.99% absorption across an extended frequency range, demonstrating enhanced resonance control and tunability. The findings indicate that more intricate square ring designs improve performance by providing enhanced resonance control and broader frequency coverage.

IV. RESULT INVESTIGATION

A. Parametric Investigations

At resonance frequencies of 0.717, 1.368, and 3.535 THz, the computed electric and magnetic field distributions are shown in Fig. 6 and Fig. 7, respectively. On the colour scale, regions of high intensity are represented by red-yellow-green areas. For the electric field strength shifts from the outermost boundary at 0.717 THz to the intermediate square at 1.368 THz (Fig. 6a–c) and finally concentrates near the innermost boundary at 3.535 THz. Similarly, the magnetic field (Fig. 7a–c) shows high intensity along the horizontal outer

boundary initially, then moves toward the intermediate square, and ultimately localizes near the innermost boundary.

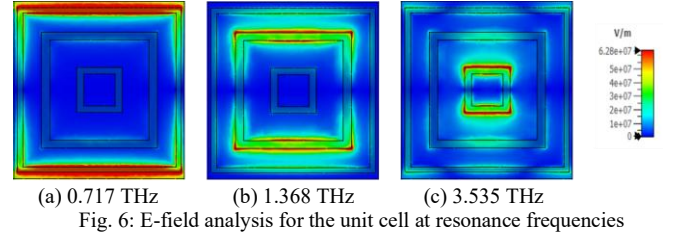


Fig. 6: E-field analysis for the unit cell at resonance frequencies

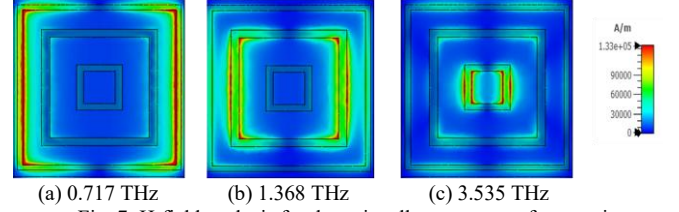


Fig. 7: H-field analysis for the unit cell at resonance frequencies

The localization of the innermost regions is the most pronounced as the frequency increases, as this pattern illustrates that both fields firmly confine near structural boundaries.

B. Performance Analysis

The study aims to assess sensibility, accuracy, and effectiveness in detecting variations in refractive index, particularly for the sensing of cancerous analytes. This Tri-band THz meta-sensing absorber exhibits strong refractive index sensing capabilities, specifically designed for the detection of cancerous analytes. Through linear regression, the sensitivities extracted from Fig. 8(a–c) for the three resonance peaks are 133 GHz/RIU, 175 GHz/RIU, and 479.5 GHz/RIU, with near-unity R^2 values (0.995, 0.9901, 0.9951), confirming strong linearity and precision. The sensor's performance was validated from Fig. 8 (d) under controlled conditions: RI variations (1.0–2.0) in 0.2 increments and a fixed analyte thickness of 0.55 μm , mimicking real-world biosensing scenarios. As the RI increases, blue shifts in resonance frequencies are observed, with absorption amplitudes exceeding 98% at lower frequencies though gradual degradation occurs at higher bands. Spectral resolution is characterized by narrow FWHM values (0.0551, 0.1077, and 0.2449), yielding high quality factors ($Q = 13.0127, 12.7019, 14.4324$) and Figs of merit ($\text{FoM} = 2.4138, 1.6249, 1.9579$). The linear resonance shifts and stable absorption ($>99\%$) underscore the sensor's reliability for detecting subtle RI changes, such as those in cancer cells.

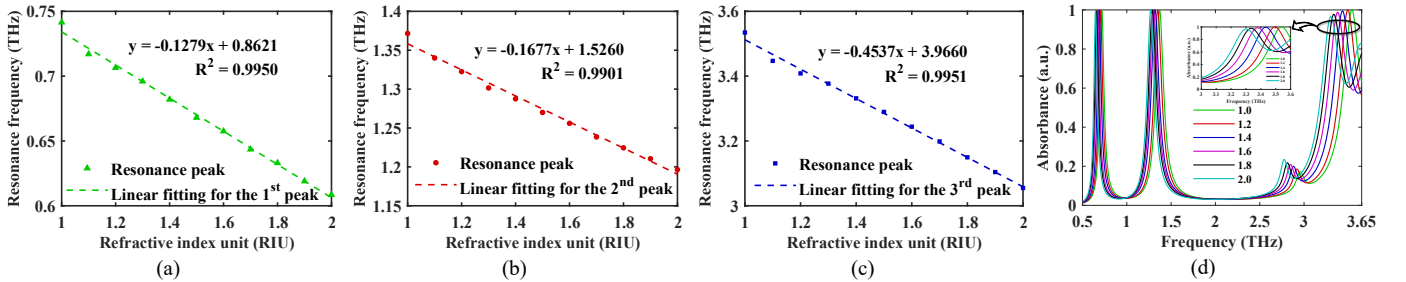


Fig. 8: Performance analysis of absorbance versus frequency in terms of refractive index (a–c), variation of refractive index (d) for optimized sensitivity of the metamaterial absorber at resonance frequencies and shift with linear fit.

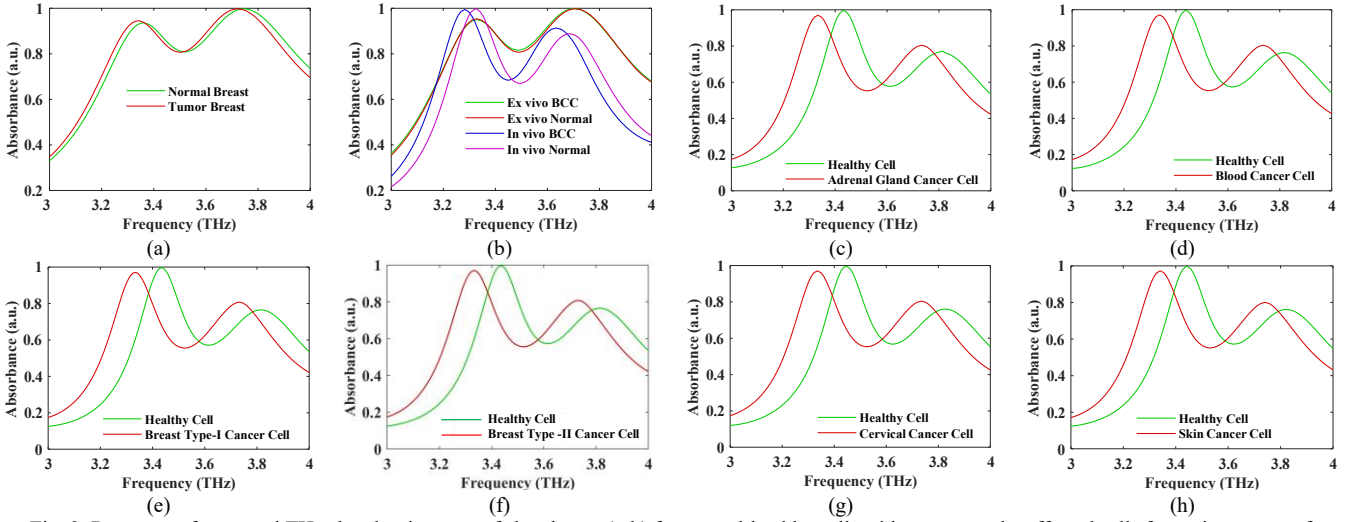


Fig. 9: Response of proposed THz absorber in terms of absorbance (a-h) for normal-healthy cells with respect to the affected cells for various types of cancer cells mentioned in Table-II.

The design provides a flexible and high-precision detection over a number of different frequency bands, with sensitivities ranging from 133 to 479.5 GHz/RIU, Q-factors exceeding 12.7 and FoM values ranging from 1.62 to 2.41. These metrics, combined with its ability to differentiate analytes via dielectric properties, position the absorber as a promising tool for biomedical diagnostics and environmental monitoring.

V. SENSING PERFORMANCE

In MPAs, frequency shifts indicate molecular and structural differences between healthy and malignant tissues. As observed in Fig. 9 (a), healthy breast tissue demonstrates a 0.9356 absorbance resonance at 3.3595 THz while, tumor tissue shows a slight redshift to 3.3385 THz (a 21 GHz transition). Fig. 9(b) delineates the dielectric response of normal versus basal cell carcinoma (BCC) skin tissues modeled via the double Debye framework, revealing pronounced spectral shifts directly correlated with malignant transformation [11]. Notably, in vivo measurements exhibit a resonance frequency redshift from 3.321 to 3.2825 THz, while ex vivo data show a subtle blueshift from 3.3245 to 3.328 THz, reflecting distinct cancer-induced dielectric alterations. These precise frequency modulations underscore the exceptional sensitivity of THz spectroscopy for non-invasive oncological diagnostics. Complementary results in Figs 9(c-h) demonstrate the biosensor's robust capability to discriminate cancerous from healthy cells through quantifiable refractive index variations, affirming its potential as a powerful, label-free platform for early-stage

cancer detection and monitoring. Higher sensitivity was observed for blood (7357.14 GHz/RIU) and breast cancer cells (7285.714 GHz/RIU), indicating a strong frequency response to small refractive index variations. In contrast, skin and cervical cancer cells showed lower sensitivity of 5100 GHz/RIU and 4625 GHz/RIU, respectively. These findings demonstrate the capacity of sensor to detect tiny variations in refractive index between malignant and healthy tissues, with particularly high performance in detecting blood and breast related malignancies. Overall, the sensor demonstrates strong potential for high-sensitivity, label-free cancer detection. Table II [12] summarizes RI values for key cancer types, including blood (Jurkat), adrenal gland (PC12), cervical (HeLa), skin (Basal Cell Carcinoma, BCC), and breast cancer subtypes (MCF-7 and MDA-MB-231, represented as Type-I and Type-II), demonstrating the efficacy of THz spectroscopy in detecting malignant changes.

TABLE II: REFRACTIVE INDEX TABLE [12]

Type		Refractive Index		Sensitivity (GHz/RIU)
Cancer	Cell	Healthy Cell	Cancer Cell	
Skin	Basal	1.360	1.380	5100
Cervical	Hela	1.368	1.392	4625
Blood	Jurkat	1.376	1.390	7357.143
Adrenal Gland	PC12	1.381	1.395	7071.429
Breast Type-I	MDA-MB-231	1.385	1.399	7285.714
Breast Type-II	MCF-7	1.387	1.401	7256.134

TABLE III: COMPARISON TABLE

Ref.	Operation range (THz)	Number of bands	Material substrate	Q-Factor	Absorbance (%)	Sensitivity (GHz/RIU)	Applications (Sensing/Detection)
[2]	0-1	2	PET	20.382, 16.286	99.77, 98.4	42, 704	Cervical Cancer
[3]	3.65-3.75	1	FR-4	92.75	99	1447	Cancer Cells
[4]	0-1.2	1	PET	12.8, 13.5	99.7, 99.6	51.5, 76	Skin Cancer
[5]	3.715-3.825	2	GaAs	34.27	97	360	Cancer Detection
[6]	4.25-4.75	3	-	55.34	96.63, 99.90, 98.84	2050	Skin, Blood, Breast, Cervical cancer
[7]	2.28-2.3	1	GaAs	47.15	99.43	502	Brain Cancer
[8]	1.06-1.15	1	Polyimide	13.2	98	556.325	Hemoglobin, Cancer cell
Proposed work	0.7-3.6	3	FR-4	13.0127, 12.7019, 14.4324	99.99, 99.95, 99.99	7357.14	Cancer Detection

VI. COMPARATIVE ANALYSIS

Comparative studies showed in Table III, that various metamaterial absorbers have been proposed for different applications such as skin, blood, breast, cervical cancer cell as well as different biological sample sensing like Hemoglobin [2–8]. These designs typically operate within specific THz ranges, with performance varying based on substrates with parameter of sensitivity, Q-factors, and absorbance properties. The proposed work demonstrates significant advancements by achieving high absorption rates and excellent sensitivity for cancer detection. Furthermore, the developed Tri-band meta-absorber has superior sensing capabilities to those of earlier investigations, highlighting its potential for sensing applications.

VII. CONCLUSION

This study outlines the creation and thorough examination of a high-performance, triple-band terahertz metamaterial absorber customized for the detection of cancer cells. The suggested configuration, constructed on an economical FR-4 substrate and featuring concentric square-ring resonators, achieves near-unity absorption with 99.99% at 0.717 THz, and 99.95% absorbance at 1.368 THz and 3.535 THz exhibiting high sensitivity to refractive index variations. Through rigorous parametric optimization using a genetic algorithm, the absorber achieves high sensitivity above 7000 GHz/RIU for blood and breast cancer tissues, with strong responses also observed in adrenal, cervical, and skin tissues, confirming its effectiveness in detecting subtle tissue variations with quality factors exceeding 12.7, enabling label-free, non-invasive cancer diagnostics. Extensive simulations validated the absorber's robustness against angular variation and mechanical bending, supporting its use in flexible and dynamic sensing environments. The spectral responses exhibited consistent shifts corresponding to variations in the refractive index of different biological analytes, including cancerous cells, confirming its effectiveness for precise, label-free detection. The proposed MTM outperforms existing designs with superior absorption, higher quality factors, and broader terahertz applicability, making it ideal for next-gen non-invasive biosensors.

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